

# What Claude Did

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We let Claude play the game and this is what he did.

## 1 It played once, naively

Fight one it did the obvious thing and hunted for a period.

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|-------|---------------------------------------------------------------------------------------------------------|
| 01:09 | “Exact period 5 locked in from index 4.”                                                                |
| 01:51 | “Period 12 and period 17 disagree on the very next beat.<br>That’s a free experiment — block and look.” |
| 02:10 | “Confirmed: period 12 from index 9.”                                                                    |
| 02:26 | “Fight 1 lost.”                                                                                         |

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Each period fitted everything seen. Period 12 was even put through a test designed to separate it from period 17, passed, and then survived twenty-eight further beats before it broke. The fight was lost at beat 60.

## 2 The tell

Then it called `reveal`, looked at the whole schedule, and wrote this:

The tape is a **mechanical (Sturmian) word**: beat  $i$  is OPEN iff  $\text{frac}(i \cdot a + b) < a$ .  
A single  $(a, b)$  pair reproduced all 120 beats with zero errors. That’s why every “period” I found held for a while then broke — irrational rotations produce long *almost*-periodic stretches (the continued-fraction convergents  $5/12, 17/41 \dots$ ) that eventually slip.

Nothing in its brief had mentioned Sturmian sequences, or irrational slopes, or that the schedules were generated at all. It was given the rules of a fight and four commands.

The second sentence is the fit. The third is the insight, and it is a different kind of thing: **it diagnosed the generator from the pattern of its own failures**, not from the data.

Look at what it had failed with. Periods 5, then 12, then 17. That monster’s slope was  $\alpha \approx 0.4142$ , and

$$\sqrt{2} - 1 = 0.41421\dots = [0; 2, 2, 2, \dots], \quad \text{convergents } \frac{1}{2}, \frac{2}{5}, \frac{5}{12}, \frac{12}{29}, \dots$$

Its first two hypotheses are the denominators of successive convergents, and  $17/41$  is the mediant of  $5/12$  and  $12/29$  — the next best rational approximation between them. It had not been finding wrong periods. It had been computing, one at a time and without realising, the continued-fraction expansion of an irrational number.

A rational schedule has one period and keeps it. A sequence of periods that each hold and then slip, with denominators climbing 5, 12, 17, 41, is what an irrational rotation looks like from inside. That is the signature, and it is visible only in the failures.

### 3 What followed

Having named the class it stopped playing and wrote a program.

For a candidate slope  $a$ , one observed beat confines the phase  $b$  to an arc of the circle. A history confines it to an intersection of arcs. So rather than choose a period, `engine.ts` keeps *the set of every  $(a, b)$  still consistent with everything seen* — and attacks only on beats where every surviving pair agrees the monster is open. Where they disagree it returns "?", and "?" means block.

That is the 01:51 idea — disagreement is a free experiment — promoted from a trick into a policy.

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|             |                        |                                                        |
|-------------|------------------------|--------------------------------------------------------|
| 03:57       | <code>engine.ts</code> | the model, 56 lines                                    |
| 04:34       | <code>live.ts</code>   | an autopilot: block twelve beats, then fit and predict |
| 04:34–07:58 |                        | <b>fight 2–10, played by the program. Nine wins</b>    |

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All nine at full health, with not one mistaken swing. For scale, on these monsters swinging every beat wins 0%, swinging only on guaranteed openings wins 0%, and a fixed table on the last four beats wins 47%.

### 4 And it measured them

It then checked the claim properly, brute-forcing  $(a, b)$  against all ten schedules — twenty thousand slopes by four hundred phases — and got zero errors on 1200 beats. It confirmed the identification three further ways, none suggested to it: complexity exactly  $m + 1$  at every window length, balance one, and gaps between open beats taking only two consecutive values. Those three properties are not evidence for a Sturmian word; they are the definition of one.

And it read off each monster's angle:

0.385, 0.414, 0.418, 0.433, 0.434, 0.455, 0.466, 0.472, 0.474, 0.476.

The generator draws its strike rate uniformly from  $[0.52, 0.62]$ , so the open rate is uniform on  $[0.38, 0.48]$ . It recovered the interval from ten samples, and marked that — correctly — as the least confident of its claims.